

ORACLE

Cloud Operations for Exadata Database Service on Exascale Infrastructure

Maintenance Operations Scenarios

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Office of CTO | EMEA Technology Engineering
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Agenda

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 - Update/Upgrade VM Guest
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 - Updating Cloud Tooling
 - Gather Troubleshooting Information

Introduction

Introduction



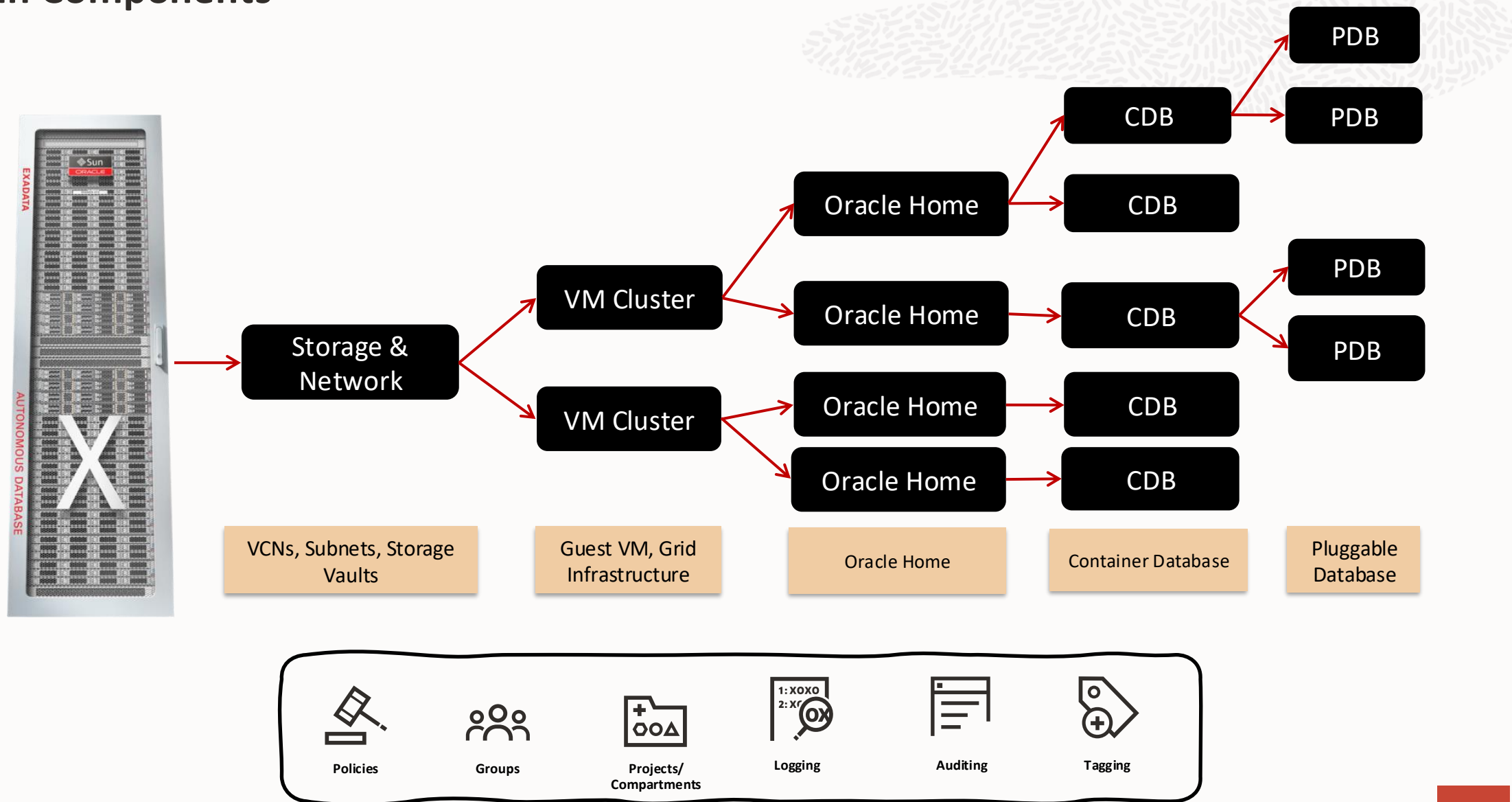
This Workshop provides information about Exadata Database Service on Exascale Infrastructure and its different components, typical operations, and maintenance activities to be performed in the platform by different automation practices

We'll cover different aspects of the infrastructure in terms of service architecture, management interfaces, the information needed to deploy a VC, OH, DBs, DG, scaling and patching

The code in this Workshop is just sample code and can be incomplete.

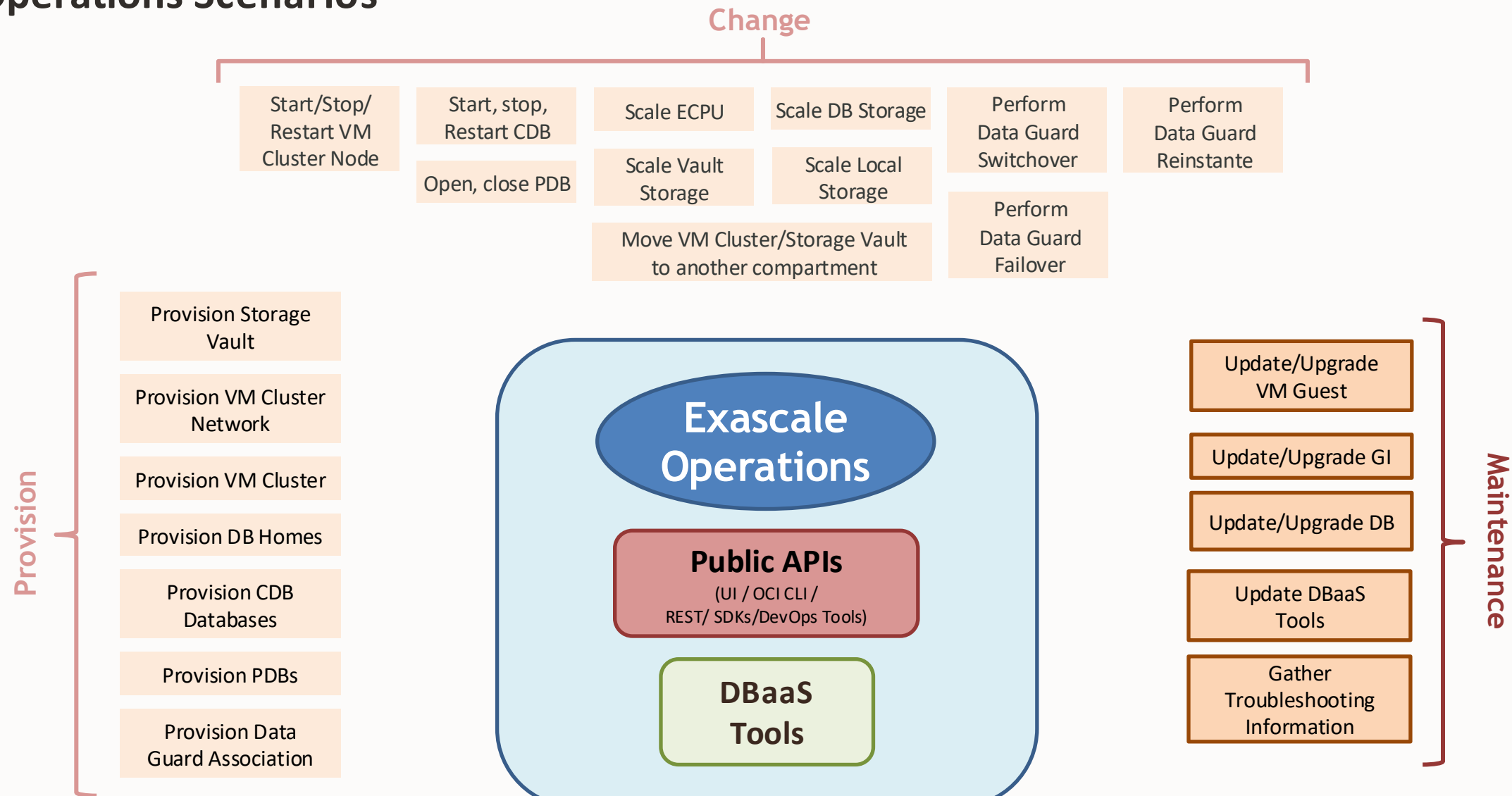
To make the sample code work in your Oracle Cloud tenancy, please replace the values for any parameters whose current values do not fit your use case and check the official documentation to include missing parameters.

Main Components



Maintenance Operations

Operations Scenarios



Update/Upgrade VM Guest

Updating an Exadata Cloud VM Cluster Operating System

Exadata VM Cluster Image Updates let you automatically update the OS image on VM cluster nodes directly from the OCI Console or APIs.

- The automated process speeds up patching, reduces errors, and removes the need for Patch Manager.
- Each patch runs a precheck to verify that the VM cluster or Database Home meets all requirements before applying updates.
 - If the precheck fails, the patch is blocked and a message is shown.
 - You can also run a standalone precheck ahead of scheduled updates.

Requirements

- Access to the OCI Object Storage service, including connectivity to the applicable Swift endpoint for Object Storage
 - Oracle recommends using a service gateway with the VCN to enable this access.
- The /u01 directory on the database host file system has at least 15 GB of free space for the execution of patching processes.
- The Oracle Clusterware is up and running on the VM cluster.
- All nodes of the VM cluster are up and running.
- Review the security rules to use with your ExaDB-XS system

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Updates (OS) → Select Actions (3 dots) on the VM OS Update → **Precheck/Apply Exadata OS Image Update**

Command Line Interface (OCI CLI)

Lists the maintenance updates

```
oci db exadb-vm-cluster list-updates --exadb-vm-cluster-id <vm-cluster-id> \  
--update-type [GI_PATCH | GI_UPGRADE | OS_UPDATE] [OPTIONS]
```

Gets information about a specified maintenance update

```
oci db exadb-vm-cluster get-update --exadb-vm-cluster-id <vm-cluster-id> \  
--update-id <update-ocid> [OPTIONS]
```

Updates the specified Exadata VM cluster on Exascale Infrastructure

```
oci db exadb-vm-cluster update --exadb-vm-cluster-id <vm-cluster-id> \  
--system-version <system-version> \  
--update-action [NON_ROLLING_APPLY | PRECHECK | ROLLBACK | ROLLING_APPLY] [OPTIONS]
```

OCI Control Plane Interfaces



OCI APIs

Lists the maintenance updates

```
GET /20160918/exadbVmClusters/<exadbVmClusterId>/updates?limit=10
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Gets information about a specified maintenance update

```
GET /20160918/exadbVmClusters/<exadbVmClusterId>/updates/<updateId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Updates the specified Exadata VM cluster on Exascale Infrastructure

```
PUT /20160918/exadbVmClusters/<exadbVmClusterId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "enabledECpuCount" : 10,
  "licenseModel" : "LICENSE_INCLUDED",
  "dbCertificateId" : "ocidl.dbcertificate.oc1."
}
```

OCI Control Plane Interfaces

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)
```

List the maintenance updates

```
list_exadb_vm_cluster_updates_response = database_client.list_exadb_vm_cluster_updates(
    exadb_vm_cluster_id="ocidl.test.oc1..<unique_ID>EXAMPLE-exadbVmClusterId-Value", update_type="GI_PATCH",
    version="EXAMPLE-version-Value", limit=234, page="EXAMPLE-page-Value", opc_request_id="1IX7KXRW1D73YVT3YWNV<unique_ID>")
print(list_exadb_vm_cluster_updates_response.data)
```

Get information about a specified maintenance update

```
get_exadb_vm_cluster_update_response = database_client.get_exadb_vm_cluster_update(
    exadb_vm_cluster_id="ocidl.test.oc1..<unique_ID>EXAMPLE-exadbVmClusterId-Value",
    update_id="ocidl.test.oc1..<unique_ID>EXAMPLE-updateId-Value", opc_request_id="P37PSR1M04KHVHINEV93<unique_ID>")
print(get_exadb_vm_cluster_update_response.data)
```

Update the specified Exadata VM cluster on Exascale Infrastructure

```
update_exadb_vm_cluster_response = database_client.update_exadb_vm_cluster(
    exadb_vm_cluster_id="ocidl.test.oc1..<unique_ID>EXAMPLE-exadbVmClusterId-Value",
    update_exadb_vm_cluster_details=oci.database.models.UpdateExadbVmClusterDetails(
        display_name="EXAMPLE-displayName-Value", total_e_cpu_count=971, enabled_e_cpu_count=380,
        vm_file_system_storage=oci.database.models.ExadbVmClusterStorageDetails(total_size_in_gbs=139),
        system_version="EXAMPLE-systemVersion-Value",
        grid_image_id="ocidl.test.oc1..<unique_ID>EXAMPLE-gridImageId-Value", update_action="NON_ROLLING_APPLY",
        ...),
    ...)
print(update_exadb_vm_cluster_response.data)
```

Update/Upgrade Grid Infrastructure

Software Images

Software images are resources within your tenancy that you create before provisioning or updating a VM cluster, Exadata Cloud Infrastructure instance, Database Home, database, or Grid Infrastructure.

- Types:
 - Grid Infrastructure
 - Database
- Procedure:
 - Create custom Database/GI images based on Software Images, RU, and one-off (interim) patches.
 - Provision a new Database/GI home with the desired RU and one-off (interim) patches.
 - Out of place Database/GI update
- Tooling Options:
 - Oracle Cloud Web based: **UI, OCI-CLI, REST API, SDK**
 - Guest VM Command Line Interface (CLI): **dbaascli**
 - IaC: **Terraform**
 - OCI Managed Services: **Exadata Fleet Update, Fleet Application Management**

Software Images



RECOMMENDED

- Ensures consistent, standardized patch levels across environments.
- Reduces patching time by deploying pre-patched images.
- Avoids manual intervention during patch apply.
- Minimizes human error during patching and configuration.
- Simplifies rollback by allowing quick switch-back to the previous Oracle Home or fast redeployment of previous images.
- Eases large-scale updates as can be used combined with other services as Exadata Fleet Update

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Updates (GI) → Select Actions (3 dots) on the DB System Patch → **Precheck/Apply Grid Infrastructure Update**

Command Line Interface (OCI CLI)

Lists the GI versions available

```
oci db database-software-image list --compartment-id <comp-id> \  
--image-type [DATABASE_IMAGE | GRID_IMAGE] [OPTIONS]
```

Updates the specified Grid Infrastructure

```
oci db exadb-vm-cluster update --exadb-vm-cluster-id <vm-cluster-id> \  
--grid-image-id <gi-version> \  
--update-action [NON_ROLLING_APPLY | PRECHECK | ROLLBACK | ROLLING_APPLY] [OPTIONS]
```


OCI Control Plane Interfaces



OCI APIs

Lists the GI Software Updates

```
GET /20160918/databaseSoftwareImages?compartmentId=<compartmentId>&limit=10
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
```

Updates the specified GI on Exadata VM cluster

```
PUT /20160918/exadbVmClusters/<exadbVmClusterId>
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "enabledECpuCount" : 10,
  "licenseModel" : "LICENSE_INCLUDED",
  "dbCertificateId" : "ocidl.dbcertificate.oc1.xxx"
  "gridImageId" : "<GI-Image-ID>"
}
```

OCI Control Plane Interfaces

Python SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)
```

List the maintenance updates

```
list_database_software_images_response = database_client.list_database_software_images(
    compartment_id="ocid1.test.oc1..<unique_ID>EXAMPLE-compartmentId-Value",
    image_type="GRID_IMAGE", image_shape_family="EXADBXS_SHAPE", ...)
print(list_database_software_images_response.data)
```

Update the specified Exadata VM cluster on Exascale Infrastructure

```
update_exadb_vm_cluster_response = database_client.update_exadb_vm_cluster(
    exadb_vm_cluster_id="ocid1.test.oc1..<unique_ID>EXAMPLE-exadbVmClusterId-Value",
    update_exadb_vm_cluster_details=oci.database.models.UpdateExadbVmClusterDetails(
        display_name="EXAMPLE-displayName-Value", total_e_cpu_count=971, enabled_e_cpu_count=380,
        vm_file_system_storage=oci.database.models.ExadbVmClusterStorageDetails(total_size_in_gbs=139),
        system_version="EXAMPLE-systemVersion-Value",
        grid_image_id="ocid1.test.oc1..<unique_ID>EXAMPLE-gridImageId-Value", update_action="NON_ROLLING_APPLY",
        ...),
    ...)
print(update_exadb_vm_cluster_response.data)
```

Local VM Command-Line Interfaces - dbaascli



Upgrade GI

```
$ dbaascli grid upgrade --targetVersion <value> [--containerURL <value>] [--executePrereqs]
  [--resume [--sessionID <value>]] [--rollback [--sessionID <value>]] [--targetHome <value>] [--softwareOnly]
  [--drainTimeoutInSeconds <value>] [--waitForCompletion <value>]
```

Update/Upgrade Database

Database patching options

***Database in-place patching** is the process of applying patches directly to an existing Oracle Database home, updating the software in that same location without creating or switching to a new Oracle home.*

***Database out-of-place patching** is the process of applying patches to a new Oracle Home and then switching the database to that updated home, leaving the original home unchanged.*

Best practices

- Back up your databases before you apply any patches.
- Patch a VM cluster before you patch the Databases Homes and databases on that resource.
- Run the precheck operation to ensure your VM cluster or Database Home meets the requirements for that patch.
- Use **out-of-place** patching if possible:
 - Less downtime
 - Easier roll back the database

OCI Control Plane Interfaces

OCI Console

Menu → Oracle AI Database → Oracle Exadata Database Service on Exascale Infrastructure → VM Clusters →
Select VM Cluster → Container databases → Select the Container database → More actions → **Move to another home / Upgrade**

Command Line Interface (OCI CLI)

Lists the DB versions available

```
oci db database-software-image list --compartment-id <comp-id> \  
--image-type [DATABASE_IMAGE | GRID_IMAGE] [OPTIONS]
```

Upgrade the specified Database

```
oci db database upgrade-with-database-software-image --database-id <db_ocid> [OPTIONS]
```

OCI Control Plane Interfaces

OCI APIs

Upgrade a Database

```
POST /20160918/databases/<databaseId>/actions/upgrade
Host: database.us-phoenix-1.oraclecloud.com
<authorization and other headers>
{
  "databaseUpgradeSourceDetails" : {
    "dbVersion" : "19.7.0.0",
    "source" : "DB_VERSION"
  },
  "action" : "PRECHECK"
}
```

Phyton SDK

```
import oci
config = oci.config.from_file()
database_client = oci.database.DatabaseClient(config)
```

Update a Database

```
update_database_response = database_client.update_database(
    database_id="ocidl.test.oc1..<unique_ID>EXAMPLE-databaseId-Value",
    update_database_details=oci.database.models.UpdateDatabaseDetails(
        db_home_id="ocidl.test.oc1..<unique_ID>EXAMPLE-dbHomeId-Value",
        new_admin_password="EXAMPLE-newAdminPassword-Value",
        old_tde_wallet_password="EXAMPLE-oldTdeWalletPassword-Value",
        new_tde_wallet_password="EXAMPLE-newTdeWalletPassword-Value",
        storage_size_details=oci.database.models.DatabaseStorageSizeDetails(
            data_storage_size_in_gbs=551,
            reco_storage_size_in_gbs=959),
        ...),
    if_match="EXAMPLE-ifMatch-Value")
print(update_database_response.data)
```

Local VM Command-Line Interfaces - dbaascli



Move database to another home

```
$ dbaascli database move --dbname <value> { --oracleHome <value> | --oracleHomeName <value> }  
  { [--resume [--sessionID <value>]] | [--rollback [--sessionID <value>]] } [--rollback]
```

Run dataPatch

```
$ dbaascli database runDatapatch -dbname <db_name> [--resume] [--sessionID] [--skipPdfs | --pdfs]  
  [--executePrereqs] [--patchList] [--skipClosedPdfs] [--rollback]
```

Upgrade Database

```
$ dbaascli database upgrade --dbname <value>  
  { --targetHome <value> | --targetHomeName <value> }  
  { [--executePrereqs] | [--rollback [--sessionID <value>]] | [--resume [--sessionID <value>]] }  
  { [--standBy] | ( [--allStandbyPrepared] [--upgradeOptions <value>] [--increaseCompatibleParameter] ) }  
  [--removeGRP] [--waitForCompletion <value>]
```


Updating Cloud Tooling

Updating Cloud Tooling

*The **cloud tooling** is automatically updated by Oracle when new releases are made available. You can update on-demand the cloud-specific tooling by downloading and applying a software package containing the updated tools.*

How can you update the cloud tooling manually?

1. Connect to a virtual machine as the opc user.

2. Start a root user command shell:

```
$ sudo -s
```

3. To show the version of latest available cloud tooling stack to update, run the following command:

```
> dbaascli admin showLatestStackVersion
```

4. To update to the latest available cloud tooling release, run the following command:

```
> dbaascli admin updateStack
```

The command takes care of updating the cloud tooling release on all the nodes of the cluster.

For more details and other available options, refer to **dbaascli admin updateStack --help**

Gather Troubleshooting Information

Local VM Command-Line Interfaces - dbaascli



Collect diagnostic log

```
$ dbaascli diag collect [--components <value>] [--startTime <value>] [--endTime <value>] [--nodes <value>] [--dbNames <value>]
{
    [--objectStoreBucketUri <value>]
    | [--destLocation <value>]
}
[--waitForCompletion <value>] [--executePrereqs]
```

Healthcheck

```
$ dbaascli diag healthcheck [--nodes <value>]
{
    [--objectStoreBucketUri <value>]
    | [--destLocation <value>]
}
```

Database Service Events

\$ as non-root
as root

Oracle Trace File Analyzer

```
#!/$ tfactl status
# tfactl start
# tfactl stop
```

sysLens

```
# systemctl status syslens
# systemctl is-enabled syslens
# /var/opt/oracle/syslens/bin/applin_ctl config_file get enable --file /etc/oracle/syslens/config/syslens.config
# tfactl get customerDiagnosticsNotifications
```

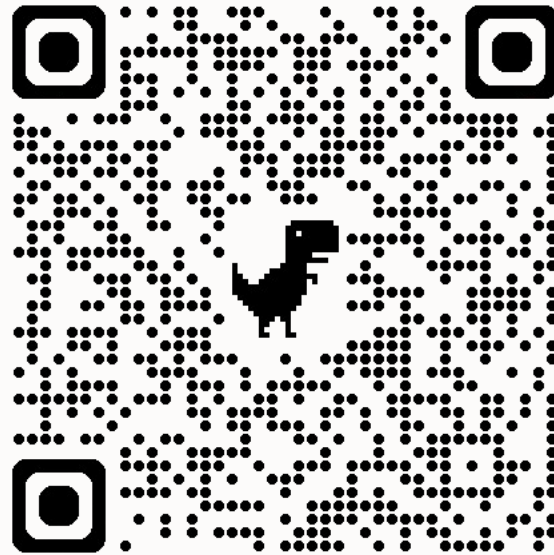
.-----.	
Hostname	
+-----+	
Configuration Parameter	Value
+-----+	
Send CEF notifications (customerDiagnosticsNotifications)	ON
'-----'	

Database Service Agent

```
#!/$ systemctl status dbcsagent.service
# systemctl start dbcsagent.service
Logs -> /opt/oracle/dcs/log/dcs-agent.log
```



Stay tuned



Operations Advisory
GitHub Repository



Thank you

ORACLE